3 **(a)** Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.

(i) State the compound that is heated with long-chain hydrocarbons to produce more useful smaller alkanes and alkenes.

..... [1]

(ii) Describe how photochemical smog is produced during the combustion of petrol in an internal combustion engine.

.....

.....

..... [2]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute HC*l*(aq).

..... [1]

(ii) Describe the thermal stability of the carbonates down Group 2.

..... [1]

(iii) Ammonium carbonate undergoes an acid–base reaction with NaOH(aq).

Explain this statement.

.....

.....

..... [2]

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(i) Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural .....

man-made .....
[2]

(ii) Write an equation to describe the role of NO_2 in the direct formation of acid rain.

..... [1]

(iii) Peroxyacetyl nitrate, PAN, is a component of photochemical smog.

Describe how PAN forms from NO_2.

.....

..... [1]

(b) A common test for nitrates is the reaction with NaOH and A*l*. Equation 1 shows the reaction.

equation 1 $3NO_3^- + 8Al + 5OH^- + 18H_2O \rightarrow 3NH_3 + 8[Al(OH)_4]^-$

(i) Deduce the oxidation state of nitrogen in NO_3^-.

..... [1]

(iii) NH_3 is a basic gas.

Describe how NH_3 is able to act as a base.

.....

..... [1]

(f) At very high temperatures, phosphorus can form P_2 molecules.

P_2 contains a triple bond, P≡P.

(i) Describe the formation of the P≡P bond in terms of orbital overlap.

.....

.....

.....

..... [2]

(ii) The bond energy of P≡P is 485 kJ mol^{-1}. The bond energy of N≡N is 944 kJ mol^{-1}. Compare the reactivity of P_2 and N_2. Explain your answer.

.....

.....

..... [1]

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..... [1]

(ii) Describe the thermal stability of the carbonates down Group 2.

..... [1]

(iii) Ammonium carbonate undergoes an acid–base reaction with NaOH(aq).

Explain this statement.

.....

.....

..... [2]

1 **(a)** **(i)** Explain the lack of reactivity of nitrogen gas, $N_2(g)$.

.....

.....

..... [2]

2 $N_2(g)$ reacts with $H_2(g)$ in the Haber process, as shown in reaction 1.

reaction 1 $\quad N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -x\ kJ\,mol^{-1}$

Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	**B**	**C**
initial molar ratio of N_2 : H_2 added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $NH_3(g)$ at equilibrium	58	x	y

(a) Describe and explain the change, if any, to the percentage yield of $NH_3(g)$ produced in **B** compared to **A**.

.....

.....

..... [1]

(d) When oxides of nitrogen escape into the atmosphere they may be involved in:

- formation of acid rain from sulfur dioxide
- formation of photochemical smog.

(i) Identify the role of NO and NO_2 in the formation of H_2SO_4 from SO_2 in the atmosphere to produce acid rain.

Use relevant equations to support your answer.

.....

.....

.....

.....

.....

..... [3]

(ii) Outline how NO and NO_2 may contribute to the formation of photochemical smog.

.....

.....

..... [2]

4 **(a)** $NH_3(g)$ reacts with $HCl(g)$ to produce $NH_4Cl(s)$, as shown.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl.

[2]

(b) An exothermic reaction occurs when $NH_4^+(aq)$ is added to $OH^-(aq)$.

(i) Identify the type of reaction.

..... [1]

(ii) Construct an ionic equation for the reaction of NH_4^+ and OH^-.

..... [1]

3 Nitrogen, N_2, is generally an unreactive molecule but it does react under certain conditions.

(a) Give **two** reasons to explain the lack of reactivity of nitrogen.

1 .....

2 .....
[2]

(b) N_2 can react with oxygen in an internal combustion engine to form a mixture of NO and NO_2. Fig. 3.1 shows a reaction scheme involving N_2.

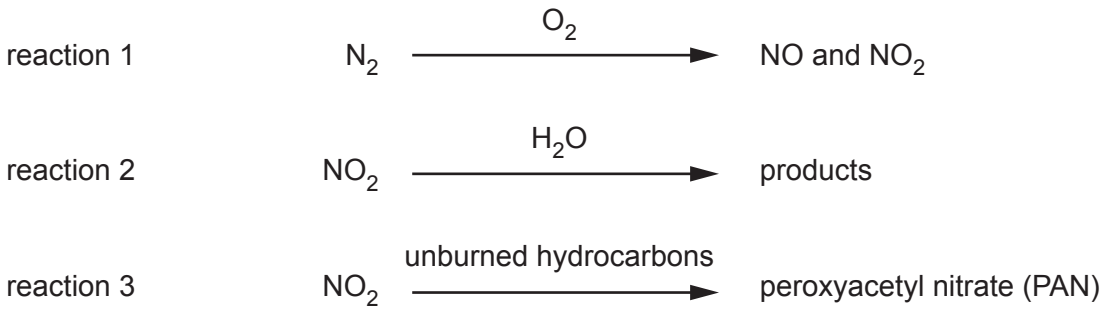


Fig. 3.1

(i) Write an equation to show the formation of a mixture of NO and NO_2 in reaction 1.

..... [1]

(ii) Give the formulae of the products of reaction 2.

..... [1]

(iii) State **one** environmental consequence of reaction 3.

..... [1]

2 NO and NO_2 react at 25 °C to give N_2O_3 as shown in the equation.

$$NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g) \qquad \Delta H = -7.2\,kJ\,mol^{-1}$$

The reaction is reversible and reaches equilibrium in a closed system.

(a) Fig. 2.1 shows how the rate of the forward reaction changes with time.

Initially, the rate of the reverse reaction is zero.

Complete Fig. 2.1 to sketch how the rate of the **reverse** reaction changes with time.

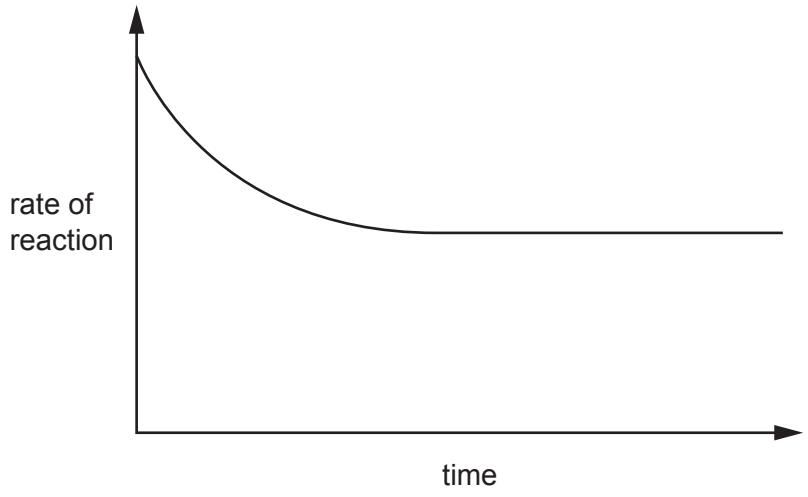


Fig. 2.1

[1]

(d) Identify one natural process and one man-made process that cause the formation of atmospheric NO and NO_2.

natural process .....

man-made process .....

[2]

(e) NO_2 is a brown gas that can be used to form nitric acid.

(i) NO_2 is a free radical.

Define free radical.

..... [1]

(ii) NO_2 has a catalytic role in the oxidation of atmospheric sulfur dioxide.

Write equations to show the catalytic role of NO_2 in this oxidation.

.....

..... [2]

(iii) State **one** environmental consequence of the oxidation of atmospheric sulfur dioxide.

..... [1]

5 **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

(i) Construct an equation for the complete combustion of **M**.

..... [2]

(b) When petrol is burned in an internal combustion engine, oxides of nitrogen are released into the atmosphere. Oxides of nitrogen are responsible for the formation of acid rain.

(i) Suggest the conditions required for the production of oxides of nitrogen during combustion of **M** in an internal combustion engine. Use an appropriate equation in your answer.

.....

.....

.....

..... [2]

(ii) Describe how acid rain is formed in the atmosphere in the presence of oxides of nitrogen and SO_2. Identify the role of the oxides of nitrogen in this process. Include **all** relevant equations.

.....

.....

.....

..... [3]

(iii) State **one** other type of air pollution that is caused by the production of oxides of nitrogen in an internal combustion engine.

..... [1]